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NeuroVision: Deep Learning-Based Brain Tumor Detection & Classification from MRI Scans

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Objective:

The main goal of our project is to build a deep learning model that can automatically detect and classify different types of brain tumors from MRI scans. We wanted to create a system that not only identifies whether a tumor is present but also recognizes its type — such as glioma, meningioma, pituitary tumor, or no tumor. By using advanced image analysis, we aim to make tumor detection more accurate, faster, and less dependent on manual inspection. We plan to use convolutional neural networks (CNNs) to train the model on a dataset of MRI images. This approach will help the model learn important visual patterns and differences in tumor shape, size, and texture, showing how deep learning can support early diagnosis and better medical decision-making, helping radiologists save time and improve the quality of healthcare outcomes. We also utilize Gradient-weighted Class Activation Mapping (Grad-CAM) to identify points in the image that signal whether a tumor is present. This gives our model the ability to give a ballpark area where the tumor may reside. Currently, MRIs, in combination with biopsies, are used to detect these types of cancers. The usage of a CNN will help to improve the rate of proper diagnosis.

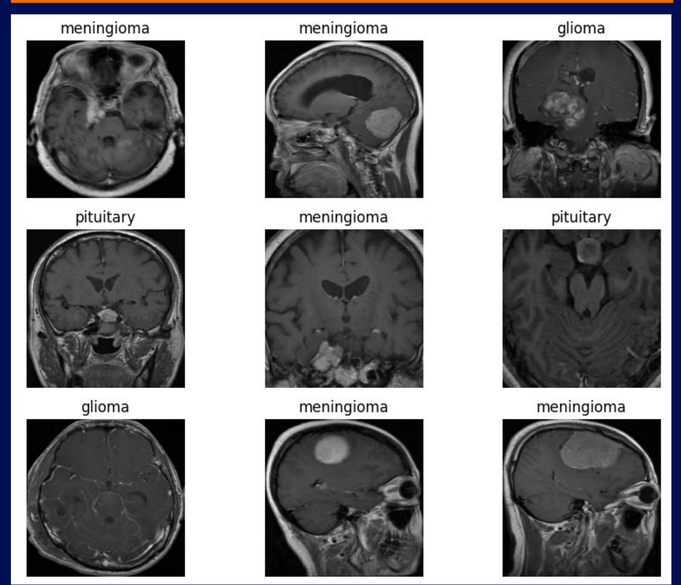

Figure 1: Sample MRIs from the Dataset

Dataset & Methodology:

The dataset consists of 7023 images of 256x256 pixel MRI scans compiled from three separate data sources, with a near-even split between the four classes (glioma, meningioma, pituitary tumor, or no tumor). The angle of the images are different as well, with some scans being different angles of the same patient, allowing for a more robust model that can better generalize.

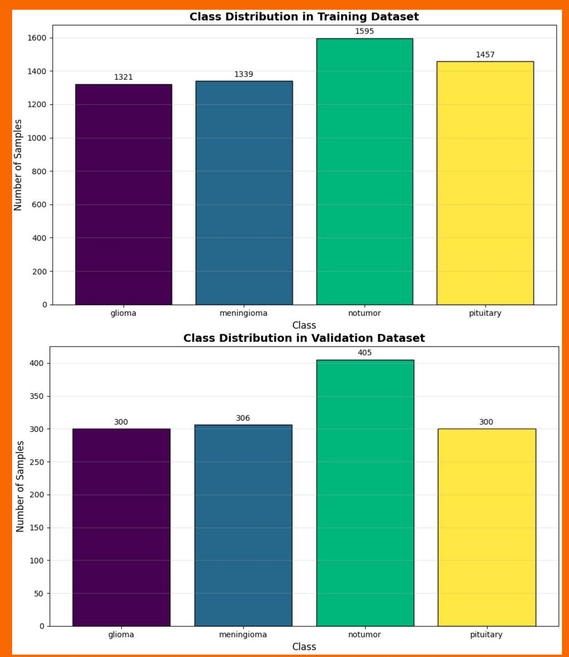

Figure 2: Class distribution in both the training & validation datasets

While the original paper utilized ResNet50 in their model, we opted to utilize EfficientNetB0 due to it's ability to be extremely accurate while also having few parameters. Looking at Figure 3, we can see that EfficientNetB0 is more accurate than ResNet50 while also having more than 15 million less parameters, resulting in a quicker, more efficient model.

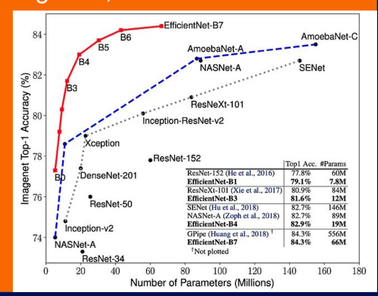

Figure 3: Graphic Showing How EfficientNet is able to retain high accuracy with a lower number of parameters (arXiv:1905.11946)

Results:

Through our model improvements using EfficientNetV2B0, a much more advanced model than the current papers had applied, we achieved an overall test accuracy of 99.7%, with 1307 of 1311 test observations being classified accurately and 0 false negatives, the most damaging type of miss. This is better than the current leading model of 99.5% on the same dataset.

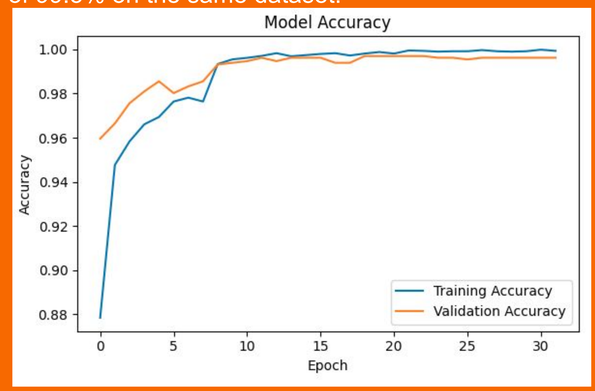

Figure 4: Model accuracy from epoch to epoch

We used a model that optimized at the 11th epoch, maxing out at a 99.95% training accuracy with a 99.69% validation accuracy upon training. Looking at the confusion matrix below, you can see there are only 4 errors, with the most frequent being a misclassification of meningioma and pituitary. Other than this error of 2 observations, the model was essentially perfectly classifying every observation.

Additionally we were able to use Grad-Cam Visualization to identify where the tumor most likely is, giving our model the ability to classify at the highest rate as well as locate at a high rate where the tumor may be. An example of this can be seen in Figure 6.

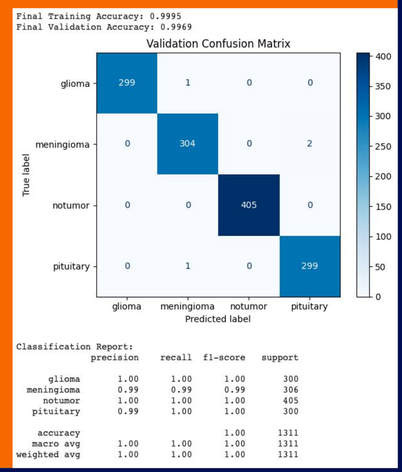

Figure 5: Model metrics & Confusion Matrix

Conclusions:

Through our model we were able to achieve the highest public accuracy of any brain tumor detection model. Using the EfficientNetV2B0 Convolutional Neural Network model, we were able to reach a test accuracy of 99.7%, surpassing published papers and all Kaggle submissions. Using a unique combination of techniques like transfer learning, learning rate reduction, early stopping, and Grad-Cam, we were able to push forward the capabilities of how accurate brain tumor detection and classification can be.

Future Work:

With an extremely accurate and precise model, a next step would be to implement it into medical software. Although brain scans already utilize machine learning to classify cancers, this model can improve upon the process. Additionally, the model can be expanded to approach a larger variety of brain cancers.

Citations

Dataset: Bhuvaji, S. (Year). *Brain Tumor Classification (MRI)*. Kaggle. https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri

Randellini, Enrico. "Image Classification: ResMet vs EfficientNet vs EfficientNet_v2 vs Compact Convolutional Transformers." *Medium*, 5 Jan. 2023, medium.com/@enrico.randellini/image-classification-resnet-vs-efficientnet-vs-efficientnet-v2-vs-compact-convolutional-c205838bbf49.

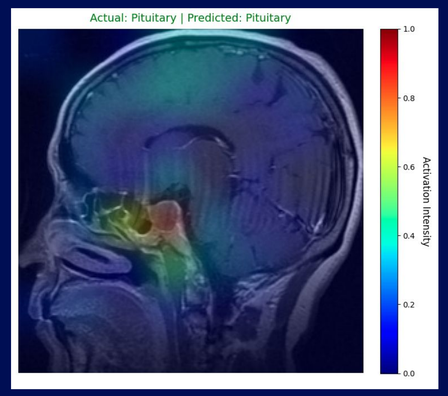

Figure 6: Grad-CAM Visualization on MRI